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Article in *Western Journal of Applied Forestry* · April 2003

DOI: 10.1093/wjaf/18.2.88

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BREEDING VALUES OF PARENTAL TREES, GENETIC WORTH OF SEED ORCHARD SEEDLOTS AND YIELDS OF IMPROVED STOCKS IN BRITISH COLUMBIA

Abstract

This report describes the procedures that are currently used in British Columbia for predicting the breeding values of parents, estimating the genetic worth of orchard seedlots, and projecting the yields of genetically improved stocks.

Breeding value is a measure of the genetic quality of an individual as a parent. There are several procedures available for its estimation/prediction. Among those, the best linear prediction (BLP) relaxes most of the assumptions required by the others and minimizes the error variance of prediction. In most situations in British Columbia, it should provide predictions with satisfactory accuracy and precision with greatly reduced computational complexity. In this province, breeding value for growth potential is expressed as percent gain of stem volume over the unimproved population at a designated rotation age.

Genetic worth is an important attribute of the genetic quality of a seedlot. It represents the average level of genetic gain expected for the trait of concern at a designated rotation age when a seedlot is used for reforestation. Currently, the genetic worth of a seedlot is estimated by the mean breeding value of all the parents, including those that contribute to pollen contamination and supplemental mass pollination, weighted by their proportional gamete contributions

The yield of a genetically improved plantation is projected by incorporating the genetic worth of the seedlot into the existing growth model developed based on extensive data from managed unimproved stands. The current approach not only takes account of the stand dynamics determined by site conditions and silvicultural regimes but also the declining nature of expected gain over time due to imperfect age-age genetic correlation.

Because of errors from genetic and environmental sampling, measurement and analysis, as well as possible violation of model assumptions, estimates/predictions may still be subject to errors and/or biases. Various conservative measures have been taken to minimize any possible upwards biases. As more matured data and advanced analytical technologies become available, both the accuracy and precision will be improved. The advancement made in the procedures described in this document should contribute to superior decisions in many aspects of forest management.

Keywords: breeding value prediction, seedlot rating, growth and yield projection.

Introduction

Tree improvement is becoming an increasingly vital component of forestry in British Columbia. Continuously shrinking forest land base, increasing demands for wood, and intensified global competition require more and better wood being produced on less land with lower cost. It has been well understood that genetic improvement is the most cost-effective approach to meet this challenge and is the best silviculture investment for the future of B.C. forestry. Besides increasing growth potential and wood quality, genetic improvement may also enhance resistance to various damaging agents such as diseases, insects, drought, and frost etc. Genetically improved stocks have demonstrated much faster early growth and, therefore, require less time for stand establishment. This may not only reduce cost on tending the young stands (e.g., brush control) but also accelerate the recovery of ecosystems.

Tree improvement practice in British Columbia was initiated in late 1960s and since then, active breeding programs have been established for all the major commercial species, including coastal and interior Douglas-fir, interior spruce, lodgepole pine, western hemlock, yellow-cedar, Sitka spruce, western red-cedar, western white pine and western larch. Many thousands of phenotypically superior trees selected from natural stands are under genetic testing from which the top performers are chosen as parents for seed production and/or advanced generation breeding. So far, 85 seed orchards have been established and more are planned. Genetically improved stocks accounted for over 30% of the total provincial planting in 1998 and this proportion is expected to reach 75% in 2007.

Given the vast investment that have been made in tree improvement and the extensive use of genetically improved stocks in operational planting in the province, it is crucial to ensure that the truly best parents are delivered through our breeding programs, the highest quality seed is produced from our seed orchards and used for reforestation, and the most appropriate management strategies are applied to the genetically improved plantations. To accomplish these tasks, the genetic merit of tested materials and seed orchard seedlots must in the first place be precisely and accurately evaluated and the growth and yields of genetically improved stocks under different environmental conditions and silvicultural regimes must be reliably modeled.

In British Columbia, breeding values (*BV*) of stem volume are estimated/predicted for all the tested parental trees and are used to quantify their genetic merit in growth potential (Xie 1997). Every seed orchard seedlot is rated for its genetic worth (*GW*) which represents the average level of gain expected from this seedlot at a designated rotation (Woods et al. 1996). The genetic worth of a seedlot is then incorporated into the existing growth and yield model developed based on extensive data from managed stands of unimproved trees to predict the yields of operational plantations established using genetically improved stocks.

This report describes the genetic concept of breeding values, details the analytical procedures employed in various breeding programs, and discusses the relationships among those different procedures. It also describes how the genetic worth of a seedlot is estimated and how it is incorporated in the existing growth model to predict the yield of the resulting plantation in British Columbia. The assumptions that are made during the development of the analytical procedures, the implications in case of their violation, and the various measures that have been taken to achieve reliable and conservative predictions are also discussed.

Genetic concept of breeding value

Breeding value (*BV*) is a measure of the genetic quality of an individual as a parent. Since an individual can only pass on its genes and not its genotype to its progeny, it is the additive component of its genotypic value that determines its genetic superiority as a parent, or its breeding value. In other words, breeding value is the additive component of the genotypic value of an individual or the value transmitted from a parent to its progeny (Falconer 1989).

When an individual is randomly mated to the other individuals in the population and all the progeny are measured in all the environmental conditions, the breeding value of this individual equals twice the mean phenotypic value of its progeny. This is because that in such situation, 1) the average non-additive genetic and environmental effects equal zero; and 2) only half the genes are contributed by the individual while the another half is a random sample from the population which have an expected mean genic value of zero. For convenience, breeding values are usually expressed as deviations from the population mean (It is even more convenient to express these deviations in percentage terms as discussed later in this paper).

Since the phenotypic value of the progeny is not only determined by the genes the individual carries, but also the genetic composition (or gene frequencies) of the population from which the mates are drawn and the environment where the progeny grow, breeding value is a joint property of the individual, the population and the target environment. It is improper to make reference to the breeding value of an individual without specifying the population in which it is to be mated and the environment where its progeny are to be grown. In British Columbia, different seed planning zones (or breeding zones) have been delineated to optimize adaptations to economic constraints and biological conditions. For all species, the breeding value of an individual in a seed planning zone is assessed by its progeny sampled from and tested in the same zone, and if selected, it is to be mated with others from this zone in a seed orchard and the seed orchard seedlings are to be deployed within the same zone. Thus, seed planning zone defines both the background population and the target environment.

Measuring all the progeny of a parent in all the environmental conditions is, however, almost (if not absolutely) impossible in any real situations. Instead, only a small

sample of progeny is tested at a limited number of sites within a target environment. With errors from both genetic and environmental sampling, the true breeding value of a parent is always unknown and it has to be predicted/estimated from the observed data using analytical procedures developed based on statistical and quantitative genetic theories.

Development of analytical procedures

In forest tree breeding, various approaches have been used to find a function of the observed data that takes into account both the genetic and environmental sampling errors to provide an estimate/prediction as close as possible to the true breeding value of a parent. They fall into four categories, namely, conventional gain estimation (Falconer 1989), Burdon's site-based selection index (Burdon 1979), best linear prediction and best linear unbiased prediction (White and Hodge 1989).

1. Conventional gain estimation

The heritability (or repeatability) of family means (h_f^2) is a well known genetic parameter. It is the ratio of family variance component to the variance of family means and expresses the degree of correspondence between the breeding value of the parent (BV) and the phenotypic value of its progeny (Falconer 1989). When breeding value is expressed as deviation of the progeny mean (y) from the population mean (α) and the progeny are half-sibs, BV has the following relationship with the general combining ability (GCA) of the parent and the selection differential (S) of the half-sib family:

$$\begin{aligned} BV1 &= 2h_f^2(y - \alpha) \\ &= 2h_f^2 GCA \\ &= 2h_f^2 S . \end{aligned}$$

BV1 is identical to the estimator of the expected gain from parental selection based on the performance of half-sib progeny (i.e., backward selection). Therefore, it is referred to as the "conventional gain estimation procedure" to distinguish it from the others. While it is easy to understand and simple to apply, this procedure has at least the following limitations:

- 1) Since the same genetic parameter, h_f^2 , is used to weight the GCA or selection differential (S) for all the parents, the degrees of correspondence between the mean phenotypic value of the progeny and the breeding value of the parent are assumed to be the same. However, they may vary among parents when the data are unbalanced.
- 2) When multiple sites are involved in the test, breeding values are usually estimated through replacing S (or GCA) in **BV1** by the arithmetic average across all sites and using the combined-site heritability estimate (h_f^2). In such case, all the sites are

equally weighted while the test precision at different sites may possibly be quite different (i.e., $h_{fi}^2 \neq h_{fj}^2$).

3) It is unable to properly deal with situations where variance structures are heterogeneous across sites even though the test precision is the same. For instance, two sites may have the same heritability but markedly different variances due to scale effects. In such situation, the same selection differential is more impressive at the test with smaller mean, and accordingly, this site should be given a larger weight.

4) It is assumed that there is no genotype-by-environment interaction or such effect is negligible.

5) It is unable to deal with situations where the assessment trait differs from the target trait and/or the test environment differs from the target environment.

A modified approach that uses the mean selection differential weighted by heritabilities ($\bar{S}_w$) rather than the arithmetic average:

$$BV2 = 2h_f^2 \bar{S}_w ,$$

where

$$\bar{S}_w = \frac{\sum_i h_{fi}^2 S_i}{\sum_i h_{fi}^2} ,$$

may be used to remove the second limitation (i.e. variation in test precision among sites), but the others still remain.

2. Burdon's site-based selection index

Burdon (1979) developed a generalized multi-trait selection index for selecting parents using half-sib progeny test information from several sites. In this procedure, observations at different sites are treated as distinct traits and a multi-trait selection index is constructed as the form:

$$I = \sum_i^s b_i (y_i - \alpha_i) ,$$

where,

I is the index value of a parent,

s is the number of test sites,

y_i is the progeny mean of the parent (or the family mean) at the i -th site,

α_i is the progeny mean of all parents (or the overall family mean) or the mean of the genetic check lots at the i -th site, and

b_i is the least-squares estimate of the coefficient (weighting factor) for S_i .

The b 's are estimated from:

$$b = P^{-1} A w ,$$

where

b is a $s \times 1$ vector of weights given to the selection differentials at the s sites,
 P is a $s \times s$ matrix of the phenotypic variances and covariances of family means,
 A is a $s \times s$ matrix of half the additive or twice the family variances and
 covariances, and
 w is a $s \times 1$ vector of the economic weights for observations at the s sites. Since
 the same trait is measured at all sites, the weights are the same and equal to 1.

The selection index can be converted to the breeding value of the parent by:

$$BV3 = I/s.$$

An alternate approach, which combines the conventional gain estimation concept with Burdon's selection index, has also been used (Carlson and Murphy 1995):

$$BV4 = 2h_f^2 \frac{I}{\sum_i b_i} .$$

Where,

$$\frac{I}{\sum_i b_i} \text{ is equivalent to the weighted mean selection differential, } \bar{S}_w, \text{ in } BV2.$$

Compared with the previous approach, Burdon's site-based selection index accounts for variable test precision as well as heterogeneity of variance and covariance structures across sites. It also has many other advantages as discussed by Burdon (1979). However, as the conventional approach does, it ignores the fact that individual test environments vary among sites and are rarely, if not never, exactly the same as the target environment. Consequently, neither the h_{fi}^2 's nor the b 's in the above two approaches are the proper measures to the degrees of correspondence between the observed phenotypic means of the progeny and the true breeding value of the parent. It is also unable to deal with the situation where the assessment trait differs from the target trait. In addition, since the same set of weighting factors are applied to all the families it has the same first limitation as described above. This procedure also requires that each family is tested at all the sites (i.e. no missing family/site subclasses), which is often not the case.

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34 **3. Best Linear Prediction (BLP):**

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Best Linear Prediction (BLP) is a method of finding a linear combination of the data that minimizes the expected value of the squared difference between the true and predicted breeding values (White and Hodge 1989). According to this method, the breeding value of a parent is predicted as:

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$$BV5 = C'V^{-1}(y - \alpha) .$$

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Where, $\mathbf{C}$ is a $s \times 1$ vector of genetic covariances between the observed half-sib family means and the breeding value being predicted. Given that the trait measured is identical to the trait for which breeding values are being predicted and the open-pollinated families are truly half-sibs, the elements in $\mathbf{C}$ are twice the type-b covariances between family means in the test and the target environments. In situations where the test environments are the same as the target environment, elements in $\mathbf{C}$ are equal to half the additive genetic variance of the population (or twice the family variance component from the combined-site analysis). However, as mentioned earlier, this rarely happens. Even in the same seed planning zone, environmental conditions at different test sites may be different and may demonstrate varying degrees of deviations from the overall environment conditions of the zone (i.e., the target environment). In our analysis, the target environment is defined as that in which the population has a mean, variance of family means, and additive genetic variance equal to those from the combined analysis across all the test sites within the seed planning zone. Since both the test and the target environments are defined by the same seed planning zone, elements in the $\mathbf{C}$ vector are estimated as

$$c_i = 2r_{Bti}\sigma_{\bar{F}_t}\sigma_{\bar{F}_i} \quad (\text{White and Hodge 1989}).$$

Where,

$$r_{Bti} = \frac{1}{s-1} \sum_{i \neq j}^s r_{Bij}, \text{ is the type-b family mean correlation between the target}$$

and the i-th test environments.

r_{Bij} is the type-b family mean correlation between tests i and j.

$\sigma_{\bar{F}_t}$ and $\sigma_{\bar{F}_i}$ are the standard deviations of family means in the target and the i-th test environments, respectively.

$\mathbf{V}$ equals $\mathbf{P}$ as defined in **BV3**.

The properties of BLP have been discussed comprehensively by White and Hodge (1989). Since both $\mathbf{V}$ and $\mathbf{C}$ can be constructed individually for each family according to the quantity and quality of the observed data, BLP overcomes the first limitation of the two previous procedures. It also automatically accounts for any types of heterogeneous variances and covariances simultaneously providing $\mathbf{C}$ and $\mathbf{V}$ are properly specified. BLP is of special advantage in providing reliable predictions of parental breeding values from messy data. It also has much broader applications than the other two procedures as the elements in $\mathbf{C}$ are genetic covariances between the observed data and the breeding value being predicted which can be for different traits, for the same trait at different ages, and/or in different target environments. Since BLP minimizes the error variance of the prediction and maximizes the correlation between the predicted and the true breeding values, it minimizes selection error and maximizes expected genetic gain.

4. Best Linear Unbiased Prediction (BLUP):

One of the assumptions under which the Best Linear Prediction (BLP) is derived is that the fixed effects (i.e., α) are known constants. In real situation, they are

never exactly known and estimates must be used. When simple estimates of the fixed effects can not be found with necessary accuracy and precision, Best Linear Unbiased Prediction (BLUP) is used. The sole difference between BLP and BLUP is in the approach of estimating the fixed effects (White and Hodge 1989). In BLP, the fixed effects are estimated using simple (arithmetic) averages of the sampled genetic entries or genetic check lots, while in BLUP they are estimated through the Generalized Least Squares (GLS). Given the relatively large sample sizes and balanced experimental designs in our first generation breeding programs, we believe simple estimates of the fixed effects are sufficiently accurate and precise. To avoid computational complexities, BLUP is not used in any of the breeding programs in B.C. and will not be subject to any further discussion in this report. However, this is not to say that BLUP will not be used in the future. As breeding advances, we may encounter situations where simple averages may no longer be acceptable as estimates of the fixed effects, and thus BLUP becomes more appropriate.

Relationship between different analytical procedures

As demonstrated above different procedures result in different functions of the observed data and therefore provide different estimates/predictions of breeding values. The essential difference between them is in the way of weighting the observed data. When data meet certain conditions (or assumptions), different procedures may generate the same weights and thus the same breeding values. To illustrate the relationship between procedures, one may consider data from two open-pollinated progeny tests with the following information:

$$y = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix},$$

$$\alpha = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix},$$

$$V = P = \begin{bmatrix} \sigma_{F1}^2 & \text{cov}_{\bar{F}} \\ \text{cov}_{\bar{F}} & \sigma_{F2}^2 \end{bmatrix},$$

$$C = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix},$$

$$A = 2 \begin{bmatrix} \sigma_{F1}^2 & \text{cov}_F \\ \text{cov}_F & \sigma_{F2}^2 \end{bmatrix}, \text{ and}$$

$$w = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Where, $\sigma_{Fi}^2 = \sigma_{Fi}^2 + \sigma_{ei}^2 = (\sigma_f^2 + \sigma_{fei}^2) + \sigma_{ei}^2$; σ_f^2 , σ_{Fi}^2 , σ_{fei}^2 and σ_{ei}^2 are the variances for the additive family effect, the effect of family at the i-th site, family by environment interaction at the i-th site, and the environment at the i-th site, respectively.

Since the expected type-b family mean covariance contains only genetic effects, $\text{cov}_{\bar{F}} = \text{cov}_F = \text{cov} = r_{B12} \sigma_{\bar{F}1} \sigma_{\bar{F}2}$ (White and Hodge 1989).

Given the fact that $(y_i - \alpha_i)$ in *BV3*, *BV4* and *BV5* equals S_i in *BV1* and *BV2*, the breeding value of a parent can be calculated using standard algebraic forms by:

$$BV1 = \frac{\sigma_f^2}{\sigma_{\bar{F}}^2} (y_1 - \alpha_1) + \frac{\sigma_f^2}{\sigma_{\bar{F}}^2} (y_2 - \alpha_2),$$

$$BV2 = \frac{2\sigma_f^2 \sigma_{F1}^2 \sigma_{F2}^2}{\sigma_{\bar{F}}^2 (\sigma_{F1}^2 \sigma_{F2}^2 + \sigma_{F2}^2 \sigma_{F1}^2)} (y_1 - \alpha_1) + \frac{2\sigma_f^2 \sigma_{F2}^2 \sigma_{F1}^2}{\sigma_{\bar{F}}^2 (\sigma_{F2}^2 \sigma_{F1}^2 + \sigma_{F1}^2 \sigma_{F2}^2)} (y_2 - \alpha_2),$$

$$BV3 = \frac{I}{s} = \frac{\sigma_{F2}^2 \sigma_{F1}^2 - \text{cov}(\text{cov} - \sigma_{F2}^2 + \sigma_{F2}^2)}{\sigma_{\bar{F}1}^2 \sigma_{\bar{F}2}^2 - \text{cov}^2} (y_1 - \alpha_1) + \frac{\sigma_{\bar{F}1}^2 \sigma_{\bar{F}2}^2 - \text{cov}(\text{cov} - \sigma_{\bar{F}1}^2 + \sigma_{\bar{F}1}^2)}{\sigma_{\bar{F}1}^2 \sigma_{\bar{F}2}^2 - \text{cov}^2} (y_2 - \alpha_2),$$

$$BV4 = \frac{2\sigma_f^2 [\sigma_{\bar{F}2}^2 \sigma_{F1}^2 - \text{cov}(\text{cov} - \sigma_{\bar{F}2}^2 + \sigma_{F2}^2)]}{\sigma_{\bar{F}}^2 [\sigma_{F2}^2 \sigma_{F1}^2 - \text{cov}(\text{cov} - \sigma_{F2}^2 + \sigma_{F2}^2) + \sigma_{\bar{F}1}^2 \sigma_{F2}^2 - \text{cov}(\text{cov} - \sigma_{\bar{F}1}^2 + \sigma_{F1}^2)]} (y_1 - \alpha_1) +$$

$$\frac{2\sigma_f^2 [\sigma_{\bar{F}1}^2 \sigma_{F2}^2 - \text{cov}(\text{cov} - \sigma_{\bar{F}1}^2 + \sigma_{F1}^2)]}{\sigma_{\bar{F}}^2 [\sigma_{F1}^2 \sigma_{F2}^2 - \text{cov}(\text{cov} - \sigma_{F1}^2 + \sigma_{F1}^2) + \sigma_{\bar{F}2}^2 \sigma_{F1}^2 - \text{cov}(\text{cov} - \sigma_{\bar{F}2}^2 + \sigma_{F2}^2)]} (y_2 - \alpha_2),$$

and

$$BV5 = \frac{\sigma_{F2}^2 c_1 - \text{cov} c_2}{\sigma_{\bar{F}1}^2 \sigma_{\bar{F}2}^2 - \text{cov}^2} (y_1 - \alpha_1) + \frac{\sigma_{\bar{F}1}^2 c_2 - \text{cov} c_1}{\sigma_{\bar{F}1}^2 \sigma_{\bar{F}2}^2 - \text{cov}^2} (y_2 - \alpha_2),$$

respectively. Where, $\sigma_{\bar{F}}^2$ is the variance of family means across sites.

It can be demonstrated that the above five functions are equal when the following conditions are all satisfied: 1), the data are balanced and all families are tested at both sites; 2), the test environments are the same as the target environment, the assessment trait is identical to the trait for which breeding value is being predicted, and the open-pollinated families are truly half-sibs; 3), the test precision at different sites is equal; 4) the effect of family-by-site interaction is negligible. In other words, when

$$\begin{aligned} c_1 &= c_2 = 2\sigma_f^2, \\ \sigma_{e1}^2 &= \sigma_{e2}^2 = \sigma_e^2 \text{ and } \sigma_F^2 = \sigma_f^2 + 1/2 \sigma_e^2, \\ \sigma_{fe1}^2 &= \sigma_{fe2}^2 = 0, \sigma_{F1}^2 = \sigma_{F2}^2 = \sigma_f^2, \text{ and } \text{cov} = \sigma_f^2, \end{aligned}$$

$$\begin{aligned} BV1 &= BV2 = BV3 = BV4 = BV5 \\ &= \frac{\sigma_f^2}{\sigma_F^2} (y_1 - \alpha_1) + \frac{\sigma_f^2}{\sigma_F^2} (y_2 - \alpha_2). \end{aligned}$$

When test precision varies between sites ($\sigma_{e1}^2 \neq \sigma_{e2}^2$), *BV1* is obviously not appreciate; *BV3* and *BV5* are still equal but they are different from *BV2* and *BV4*:

$$BV2 = \frac{2\sigma_f^2 \sigma_{F2}^2}{\sigma_F^2 (\sigma_{F1}^2 + \sigma_{F2}^2)} (y_1 - \alpha_1) + \frac{2\sigma_f^2 \sigma_{F1}^2}{\sigma_F^2 (\sigma_{F1}^2 + \sigma_{F2}^2)} (y_2 - \alpha_2),$$

$$BV3 = BV5 = \frac{2\sigma_f^2}{\sigma_{F1}^2 + \sigma_f^2 (\sigma_{e1}^2 / \sigma_{e2}^2)} (y_1 - \alpha_1) + \frac{2\sigma_f^2}{\sigma_{F2}^2 + \sigma_f^2 (\sigma_{e2}^2 / \sigma_{e1}^2)} (y_2 - \alpha_2).$$

$$BV4 = \frac{2\sigma_f^2 \sigma_{e2}^2}{\sigma_F^2 (\sigma_{e1}^2 + \sigma_{e2}^2)} (y_1 - \alpha_1) + \frac{2\sigma_f^2 \sigma_{e1}^2}{\sigma_F^2 (\sigma_{e1}^2 + \sigma_{e2}^2)} (y_2 - \alpha_2).$$

Extending to situations with more than two sites, *BV1*, *BV2*, *BV3*, *BV4* and *BV5* have the following relationships:

1), when all the second moments are homogeneous across sites, that is, the diagonal elements of *V* (or *P*) are the same, the off-diagonal elements are identical, all the elements of *C* and *A* are the same and equal to twice the off-diagonal elements of *V* (or *P*), they are equal.

2), when test precision varies between sites, that is, the diagonal elements of *V* (or *P*) are different, *BV3* and *BV5* are still equal, but *BV1*, *BV2* and *BV4* are different.

3), when the other second moments are also heterogeneous across sites, *BV1*, *BV2*, *BV3*, *BV4* and *BV5* are all different.

The magnitude of differences between breeding values derived from different analytical procedures depends on the uniformity of the observed data: the more uniform,

the less different. To illustrate these similarities (or differences) with real data, we use the 20-year test results of interior spruce height growth in the east Kootenay seed planning zone as an example. The open-pollinated progeny trials were established using a randomized complete block design with 2 to 4 replications of 10-tree row plots. The estimated phenotypic and additive genetic variance-covariance matrices (V and A), genetic covariances between the observed half-sib family means and the true breeding values (C), and family heritabilities are given below:

$$V = P = \begin{bmatrix} 2454.4 & 1383.2 & 1143.1 & 1753.3 & 1471.6 & 1043.5 & 1311.9 & 1332.6 & 1370.6 \\ 1383.2 & 1916.6 & 943.3 & 1464.9 & 1099.3 & 840.3 & 795.8 & 1251.4 & 1140.0 \\ 1143.1 & 943.3 & 1345.9 & 1252.0 & 915.2 & 727.4 & 979.5 & 982.5 & 974.0 \\ 1753.3 & 1464.9 & 1252.0 & 2740.3 & 1701.3 & 931.0 & 1215.4 & 1503.8 & 1583.3 \\ 1471.6 & 1099.3 & 915.2 & 1701.3 & 2692.9 & 891.3 & 1254.6 & 887.2 & 921.0 \\ 1043.5 & 840.3 & 727.4 & 931.0 & 891.3 & 1551.4 & 914.3 & 704.4 & 887.3 \\ 1311.9 & 795.8 & 979.5 & 1215.4 & 1254.6 & 914.3 & 4732.0 & 1020.7 & 1020.6 \\ 1332.6 & 1251.4 & 982.5 & 1503.8 & 887.2 & 704.4 & 1020.7 & 1531.3 & 1270.5 \\ 1370.6 & 1140.0 & 974.0 & 1583.3 & 921.0 & 887.3 & 1020.6 & 1270.5 & 2600.2 \end{bmatrix},$$

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$$A = 2 \times \begin{bmatrix} 1719.8 & 1383.2 & 1143.1 & 1753.3 & 1471.6 & 1043.5 & 1311.9 & 1332.6 & 1370.6 \\ 1383.2 & 1119.0 & 943.3 & 1464.9 & 1099.3 & 840.3 & 795.8 & 1251.4 & 1140.0 \\ 1143.1 & 943.3 & 444.1 & 1252.0 & 915.2 & 727.4 & 979.5 & 982.5 & 974.0 \\ 1753.3 & 1464.9 & 1252.0 & 1150.9 & 1701.3 & 931.0 & 1215.4 & 1503.8 & 1583.3 \\ 1471.6 & 1099.3 & 915.2 & 1701.3 & 1454.2 & 891.3 & 1254.6 & 887.2 & 921.0 \\ 1043.5 & 840.3 & 727.4 & 931.0 & 891.3 & 837.7 & 914.3 & 704.4 & 887.3 \\ 1311.9 & 795.8 & 979.5 & 1215.4 & 1254.6 & 914.3 & 1230.3 & 1020.7 & 1020.6 \\ 1332.6 & 1251.4 & 982.5 & 1503.8 & 997.2 & 704.4 & 1020.7 & 1128.4 & 1270.5 \\ 1370.6 & 1140.0 & 974.0 & 1583.3 & 921.0 & 887.3 & 1020.6 & 1270.5 & 1924.0 \end{bmatrix},$$

13

$$C = \begin{bmatrix} 1837.8 \\ 1511.2 \\ 1296.9 \\ 1960.6 \\ 1541.6 \\ 1143.0 \\ 1495.5 \\ 1476.9 \\ 1575.1 \end{bmatrix},$$

14

1

2

$$h_{fi}^2 = \begin{bmatrix} 0.7 \\ 0.57 \\ 0.38 \\ 0.64 \\ 0.52 \\ 0.49 \\ 0.31 \\ 0.74 \\ 0.46 \end{bmatrix},$$

3

4

$$h_f^2 = 0.81.$$

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The breeding values and their ranks derived from different analytical procedures are presented in Table 1. To compare the effects of different procedures on selection and gain estimation we may consider selecting the top 25 parents as an example. As indicated in Table 1, at least 23 of the 25 selected parents would be the same regardless the analytical procedures used to derive the breeding values. The effect on the expected gain is also very minimal. Assuming BLP is the best procedure for predicting the true breeding values of the parents, and therefore the genetic gain from selection, selection based on breeding values derived from other procedures rather than BLP (i.e., $BV5$) would only result in a loss of gain less than 0.5% ($-\Delta Gi = (1 - (\overline{BV_i} - \overline{BV5}) / \overline{BV5}) \times 100\%$). Amongst the five procedures, BLP generated the lowest predicted breeding values, as expected. In other words, other procedures tend to overestimate parental breeding values and the expected gain.

Due to objective and/or subjective reasons, various procedures have been used in different breeding programs in British Columbia. Although this should not affect the relative ranking of genetic quality of seedlots from different seed orchards since breeding values of all the parents for a species are derived using the same procedure, expected gains might have been overestimated for species in which procedures other than BLP are used. Due to its superiority as discussed above, BLP will be used as a standard procedure for predicting parental breeding values across all the breeding programs in B.C. at least in the first generation of breeding.

Converting height breeding value at assessment age to percent volume gain at rotation

Breeding values directly derived from the above procedures represent expected gain over the test population for the measured trait at the assessment age in the measurement unit. For instance, that a parent has a breeding value of 100 cm on 10-year

height can be interpreted as if this parent is crossed with other parents of equal breeding value, its offspring are expected to be 100 cm taller than average when they are 10 years old. It is more convenient and appropriate to express breeding value of height as a percentage of volume gain over the unimproved population mean at rotation. This is because 1) genetic gain is most often expressed in percentage rather than absolute terms, 2) the test population mean is often an overestimate of the unimproved population mean, and such, may result in underestimated breeding value or expected gain, and 3) our ultimate interest is gain in harvestable timber at rotation. Converting breeding value of height at assessment age to its counterpart at rotation age can be accomplished by multiplying the coefficient of additive genetic correlation between the assessment (or juvenile) and rotation (or mature) ages ($r_A(j,m)$) and assuming the coefficients of additive genetic variation at the two ages are equal (Falconer 1989). As data are not available to estimate $r_A(j,m)$, extrapolations from the Lambeth (1980) model are used (i.e., $r_A(j,m) = 1.02 + 0.308LAR$, where, LAR is the natural logarithm of the ratio of the assessment age to the rotation age). The assumption implied in using the multiplier of $r_A(j,m)$ which decreases in value as the gap between the assessment and rotation ages increases is due to the expectation that our initial selections may not have all the genes that affect later height growth and the difference increases as trees grow. Thus, $r_A(j,m)$ represents a risk factor in making selection at early ages. Since the rotation age of a species varies with site quality, a designated rotation age determined by the average age when MAI (Mean Annual Increment) reaches its maximum at many sites with various qualities for the species (80 years for interior spruce and 60 years for all others) is used.

To convert breeding values of height to stem volume, a multiplier of 2 is applied as recommended by the growth and yield scientists in the Research Branch of B.C. Forest Services. In most situations examined in the progeny test data, this appears to be a conservative multiplier; however, it tends to be more reflective of the relationship between height and volume observed in unimproved matured stands used for developing growth models in B.C. Finally, when there are no genetic controls (i.e., samples from local wild stands) in the test, and therefore, the test population mean (M_{test}) is used and the parents are superior phenotypes, a 2% "lift" is introduced to account for the expected gain from "plus tree" selection. On the other hand, when the local wild stand control mean ($M_{control}$) is used, the 2% is not added as it is incorporated in the selection differential ($y - \alpha$) by default. Therefore, the final expression of the parental breeding values on growth is percent volume gains over the unimproved population at a designated rotation age:

$$BV_{vol}(\%) = \left[\left(\frac{BV_{ht}}{M_{test}} \times 100 \right) \times r_A(j,m) \right] \times 2 + 2, \text{ or}$$

$$BV_{vol}(\%) = \left[\left(\frac{BV_{ht}}{M_{control}} \times 100 \right) \times r_A(j,m) \right] \times 2.$$

In calculating breeding values for wood density, the age-age additive genetic correlation coefficient is not introduced since we are taking the view that the critical wood in a log is the juvenile wood which is directly measured. Thus, breeding values of wood density simply reflect expected percent gain at the assessment age.

Genetic worth of seed orchard seedlots

Genetic worth (GW) is an important attribute of the genetic quality of a seed orchard seedlot (Woods et. al. 1996). It represents the average level of gain expected for the trait of concern at a designated rotation age when a seedlot is used for reforestation within its proposed seed planning zone with proper silviculture practices. In British Columbia, all the seed orchard seeds are rated for their genetic worth (GW) and reported in SPAR (the Seed Planning and Registration System) which provides ministry and non-ministry staff with an on-line access to current information on seed and vegetative lots and an on-line facility for entering seedling requests.

Until late 90s, B.C. seed orchards (regardless of species) were classified into three categories according to their makeup, and seedlots from all the orchards of the same category were assigned the same genetic worth. For stem volume, a genetic worth of 2, 5, and 8% were assigned to seedlots from orchards which were constructed using phenotypically superior wild-stand selections (i.e., first-generation untested seed orchards), using selections based on open-pollinated progeny test information (i.e., so-called 1.5 generation seed orchards, or tested first-generation seed orchards), and using selections in progeny tests from pedigreed materials (i.e., second generation seed orchards, or more correctly, advanced generation orchards since the orchard parents can be a combination of F1 and F2 selections), respectively. Assigning genetic worth in such a way is no longer considered appropriate since different species and different populations (as defined by seed planning zones) within a species may differ widely in genetic progress at a given improvement stage. Even for a given seed orchard, different seedlots may have different genetic worth. This is because the genetic worth of a seed orchard seedlot is not only determined by the genetic superiority (or breeding values) of the orchard parents but also other factors such as the proportional gamete contributions of various orchard clones and the extent and quality of pollen from non-orchard sources (e.g., contamination and supplemental mass pollination) which usually vary from seedlot to seedlot. In addition, without a more quantitative system in place to rank seed orchard seedlots, there are no real measurable incentives (value of seed) in place to produce better crops.

With information on parental breeding values, the genetic worth of a seedlot is currently estimated by the mean breeding value of all the parents (including non-orchard parents such as those that contribute to pollen contamination and supplemental mass pollination) weighted by their proportional gamete contributions. Protocols for assessing relative gamete contribution from different sources have been well defined (Woods et al. 1996).

Predicting yields of improved stocks

Genetically improved stocks are expected to produce higher yields, as they do in agricultural improvement programs. For proper timber supply projection, the gain from tree improvement should be incorporated in growth and yield analysis. Historically, genetic worth values were used as straight multipliers to yield estimates from managed stand yield curves. Predicting yield in such a way is appropriate only if the stand is harvested at the designated rotation age. Because of the expected imperfect age-age genetic correlation mentioned earlier, harvesting prior to or after the designated rotation age may under- or overestimate the yield.

A more recent approach is to translate GW into an increase in site index at the designated rotation age and then increase the site productivity by this amount on all subsequent TIPSYS (Table Interpolation Program for Stand Yields) runs with this species. As a result, the percent gain in volume is elevated initially, decreases to the prescribed level at the designated rotation age, continues to decline for a certain period and then remains roughly constant in the following years. Although the expected percent gain decreases as the plantation ages, such age-age relationship is primarily due to stand dynamics and scale effects over time (the denominator in the percent gain equation, i.e., the unimproved population mean, increases faster than the numerator, i.e., the absolute gain). Decreases in gain due to differences in genes that control growth at different ages (i.e., imperfect age-age genetic correlation) are not accounted for. Consequently, it overestimates the yield of the plantations that have past the designated rotation age, and the older the plantations are, the greater the overestimation is. It is therefore necessary to take age-age genetic correlation into consideration in predicting the yields of genetically improved plantations. The following provides a more detailed description of the current approach of incorporating genetic worth in growth and yield analysis in B.C.

With information on the genetic worth (GW) of the improved planting stock, TIPSYS first converts the expected percent volume gain at the designated rotation age (Ad) back to the percent height gain (GHs) at the selection age (As) as follows:

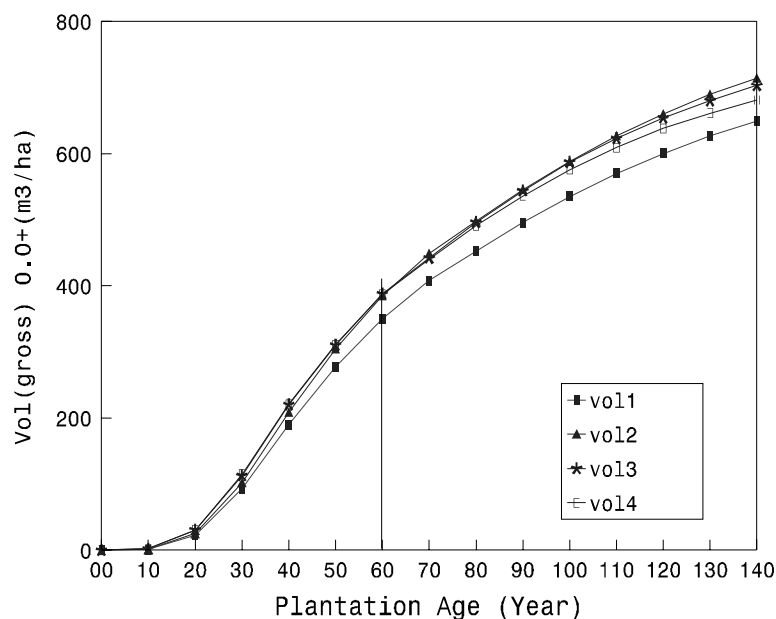
$$GHs = (GW/2) / [1.02 + 0.308 \ln(As/Ad)].$$

Then, it calculates the percent height gain (GHr) at a requested age (Ar) as

$$GHr = GHs \times [1.02 + 0.308 \ln(As/Ar)],$$

and this is added to the “base top height” of the unimproved stand (which is a function of the age and site index) as the expected height of the improved stand at the same requested age. Finally, TIPSYS applies this expected height to obtain the corresponding expected volume from its internal yield tables.

Figure 1 illustrates the predicted yields at different ages for a interior lodgepole pine plantation established with one year old stock at a space of 1100 trees per hectare. The genetic worth of the planting stock is 10% and the site index is 20. It is clear that using genetic worth as a straight multiplier or as an increase in site index tends to under and over predict yield at ages prior to and after the designated rotation.



Vol1: volume of unimproved stand;

Vol2: volume of improved stand estimated using GW as a straight multiplier;

Vol3: volume of improved stand estimated using GW as an increase of site index;

Vol4: volume of improved stand estimated using the current approach.

Figure 1. Yields predicted from TIPSy using different approaches of incorporating the genetic worth of the seedlot

General discussion

All the analytical procedures employed for estimating/predicting parental breeding values are developed based on certain assumptions. Violation of any model assumptions may not only affect the magnitude of the estimates/predictions but possibly their relative ranking, and thus, may cause selection errors. Among the procedures investigated in this report, BLP relaxes most of the assumptions and makes the “best” use of the data (i.e., minimizes the error variance of the prediction, White and Hodge 1989). Since the first moments (i.e., means of the fixed effects) can be well estimated with simple averages of the sampled genetic entries in our current testing programs (first generation), BLP should provide predictions with satisfactory accuracy and precision while greatly reducing computational complexity.

Several additional assumptions have to be made while expressing breeding value of height at some assessment age as percent gain in stem volume over unimproved population at the designated rotation age. Violation of those assumptions may result in biases of breeding value predictions but will not change the relative ranking of the parents. In other words, the expected gain may be under- or over-estimated but the realized gain will be the same when parents are selected based on their breeding values.

The first assumption that we make is that the genetic correlation of heights at two ages can be extrapolated from the Lambeth (1980) model. Although the Lambeth model was developed based on estimates of phenotypic correlation and genetic correlation may not be always linear with LAR (King and Burdon 1991), it has been used to estimate genetic correlation for growth traits by many other investigators. Empirical data from various species have demonstrated that estimated genetic correlation is typically stronger than that extrapolated from the Lambeth model (Lambeth *et al.* 1983, Riemenschneider 1988, Hodge and White 1992, Xie and Ying 1996, Gwaze *et al.* 1997). In other words, Lambeth model tends to provide conservative estimate of the age-age genetic correlation of height. Ideally, genetic correlation should be estimated separately for each species, age combination and target environment. In the absence of experiments, however, adopting the Lambeth model for extrapolating genetic correlation to rotation age is one of the best approaches available to us (Burdon 1988).

Secondly, breeding values of stem volume are converted from breeding values of height by applying a multiplier of 2, which is also considered to be conservative for estimating volume gain from selection. In practice, a multi-trait index selection approach is always used either formally or informally. Traits like stem form, crown size, branch angle and size, and others (e.g., pest resistance) also contribute to the evaluation of a parent. Gains in those traits should ultimately translate to gain in stem volume. Recently published data in *Pinus radiata* in New Zealand (Carson *et al.* 1999) indicate that percent volume gain from open-pollinated seed orchard seed is over three times higher than the corresponding height gain. Nevertheless, this conversion does not have any effect on projecting the yield of genetically improved stocks according to the current approach used

for incorporating genetic gain into growth and yield analysis (Height gain rather than volume gain is used).

The assignment of 2 percent volume gain from “plus tree” selection should be very reasonable, if not under estimated. The removal of inbreeding in seed orchards alone may well account for such gain. The breeding value of a parent estimated from its progeny that is produced by complete outcrossing (BV_{out}) has the following relationship with that estimated if the progeny is produced by partial inbreeding (BV_{in}):

$$BV_{in} = BV_{out} \times \frac{1}{1 + F} ,$$

or

$$BV_{in} = BV_{out} \times \frac{1 + t}{2} .$$

Where,

$$F = \frac{1 - t}{1 + t} \text{ is the inbreeding coefficient and } t \text{ is the outcrossing rate.}$$

According to this relationship, the removal of 5% inbreeding is equivalent to adding 2.5% gain, while an average of 10% inbreeding has been reported for natural populations of many conifer species.

It is our intention to take these conservative measures to offset the effect of possible overestimation. There are at least three factors that can potentially cause overestimation of parental breeding values. First, as progeny tests age, competition among different families may inflate genetic variance and therefore parental breeding values (Magnussen 1989). Although most of our selections were made before significant competitive effects arose, it is unlikely that we have completely unbiased results in all our tests. Secondly, our breeding values have been predicted from trials established with multi-tree-plot (4 to 10 trees per plot) designs which also tend to inflate family differences due to cohorts sampling similar environments for members of the same family (Magnussen 1993). Many of our later experiments (say since 1995) are composed of single-tree plots, which should remove such effect entirely. Finally, given the fact that the deployed superior genotypes will grow in a more balanced competitive environment in plantations than in progeny tests where both superior and inferior genotypes grow intermixed, expectations of genetic improvement based on individual tree performance as measured in progeny tests will have to be adjusted, over time, for performance on a per unit area basis.

Since the genetic worth of a seedlot (GW) is estimated as the weighted mean of parental breeding values (BV), those factors that affect the accuracy and precision of BV prediction should also influence the reliability of GW estimates. In addition, because parents are tested at a limited number of sites and breeding values are predicted for a designated rotation age based on performance at a much younger assessment age, it must be kept in mind that:

- 1) The same seedlot may generate different gains at sites with different qualities (or site indices) and/or silvicultural treatments.
- 2) Different levels of gain are expected from a seedlot when the plantation is harvested at different ages.
- 3) Two seedlots that have the same GW may generate different levels of gain if their parents were selected at different ages and the harvesting age differs from the designated rotation age.

To unequivocally project yield of an operational plantation established with genetically improved materials, one needs to know the performance of such genetically improved stock over an entire rotation in the operational plantation environment. While long term large block-plot genetic trails (e.g., realized genetic gain trails) are still too young to provide reliable information, several efforts have been made in attempting to understand the stand dynamics of genetically improved plantations and to use growth and yield data from tree improvement trials to calibrate pre-existing models developed based on unimproved trees for predicting the yield of operational plantations with genetically improved stocks. Through examining the growth trends of loblolly pine in provenance and open-pollinated progeny trials as well as operational plantations, Buford (1986) and Buford and Burkhart (1987) found that only the levels but not the shapes of the height-age and height-diameter curves differed by seed source or family and the yield equations fitted to the data from operationally planted improved and unimproved stocks were essentially the same. They suggested to use a single yield model for both improved and unimproved stocks and an increase in site index to account for increased growth from genetic improvement.

As mentioned earlier, Carson et al. (1999) analyzed data from half rotation genetic-gain trials of radiata pine in New Zealand. They found that the change in the rate of increase of basal area was much greater than the change in the rate of increase in mean top height, making increases in basal area and volume growth much greater than would be predicted from increasing site index alone. They thus indicated that additional growth in basal area over the corresponding increase in height growth must be taken into account when predicting yield of genetically improved stocks in radiata pine in New Zealand. It is important to note, however, that diameter has been the main selection trait for improved growth potential in New Zealand tree breeding programs.

In British Columbia, selection for timber productivity is primarily based on height, and therefore, the height-diameter or height-volume relationship of the selected trees is not expected to be significantly different from the unimproved stocks under the same stand condition. In other words, increases in height from genetic improvement can be reasonably (and may be conservatively) converted to volume gain using the height-volume relationships observed in the unimproved trees. Accurate prediction of height gain is then the key for accurately projecting volume gain of genetically improved stocks. Predicting yield of genetically improved stocks at different ages from a single curve with an increase in site index, corresponding to the predicted level of genetic improvement in height at a single reference age as quantified by the GW of a seedlot, is not considered appropriate.

1 This is because it ignores the fact that the expected genetic gain declines over time due to
2 imperfect age-age genetic correlation. In the approach currently employed in British
3 Columbia, gain in top height of a plantation at a given age is determined by the GW of the
4 seedlot and the genetic correlation between the selection and the given ages. Therefore,
5 the declining nature of gain over time due to imperfect age-age genetic correlation is
6 accounted for.

7
8 Despite all the improvement in parental breeding value prediction, seedlot genetic
9 worth estimation, and the incorporation of genetic gain in growth modeling, errors and/or
10 biases in predicting yields of genetically improved stocks are still unavoidable. As Carson
11 et. al. (1999) indicated, however, the error introduced into prediction models by adding a
12 genetic component is expected to be well within the acceptable limits, especially in view of
13 genetic effects being so much smaller than those of site and silviculture treatments. After
14 taken the various conservative measures as described above, predictions from the new
15 procedures should be closer to the “safe” (versus over prediction) side than they were
16 before. Both the biases and errors of prediction will be further reduced, although never be
17 eliminated, as more advanced analytical technologies and matured experimental data
18 (particularly, those from the realized gain trials) become available.

19
20 In British Columbia, the area of land being reforested with genetically improved
21 stocks is expanding at an increasing pace, particularly with the enforcement of the
22 Silviculture Practices Regulations (Part 2, Division 1) of the Forest Practices Code Act.
23 These regulations require the available seed sources of the highest genetic quality be used
24 wherever possible on Crown land which accounts for over 96% (48 million hectares) of
25 the total productive forest land in B.C. (49.9 million hectares). To efficiently manage these
26 new plantations, the stand dynamics of the genetically improved stocks has to be known or
27 reliably predicted (Goulding 1994). The advancement made in the analytical procedures
28 described in this document may not only enhance the efficiency of tree breeding and seed
29 orchard management, but contribute to superior decisions in various aspects of forestry
30 operation.

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- 5

1 **Table 1. Parental breeding values and their ranks derived from different analytical**
 2 **Procedures¹**

PARENT I.D.	BV1	RANK	BV2	RANK	BV3	RANK	BV4	RANK	BV5	RANK
70	139	2	144.1	1	157.9	2	136.8	2	107.3	1
29	149	1	138.5	2	167.2	1	144.8	1	102	2
28	126.7	4	130.2	3	146.2	4	126.6	4	98.3	3
79	118.3	6	121.3	6	142.4	5	123.4	5	92	4
30	129.5	3	127.8	4	154.7	3	134.1	3	88.3	5
48	115.8	7	123	5	128.8	7	111.6	7	85.1	6
75	119	5	110.8	7	116.5	11	101	11	85	7
49	91.4	15	95.8	13	123.7	9	107.1	9	77	8
40	108.2	8	104.5	8	126.8	8	109.9	8	76.3	9
76	105.2	9	96.1	12	103.1	17	89.3	17	74.6	10
62	95.2	13	102.9	9	122.9	10	106.5	10	74.1	11
55	98.7	12	100.6	10	107.6	12	93.2	12	70.5	12
71	94.3	14	92.8	14	105.2	15	91.2	15	69.8	13
31	99.3	11	90.3	15	100.1	18	86.7	18	69.4	14
78	100.4	10	97.8	11	130.6	6	113.2	6	68.1	15
132	68.4	22	78.5	17	103.7	16	89.8	16	67.3	16
2	75.1	19	88.9	16	106.2	13	92	13	65.2	17
7	80.7	17	76	19	105.3	14	91.2	14	57.1	18
9	84	16	76.6	18	86.9	20	75.3	20	53.3	19
12	61.2	23	70.7	20	99.6	19	86.3	19	53.2	20
81	76.5	18	69.6	22	82.8	21	71.7	21	52	21
27	58.2	24	68.2	23	78.1	22	67.7	22	48.7	22
34	74.9	20	70.5	21	76.6	24	66.4	24	48.3	23
20	53	26	57.3	27	65.7	28	56.9	28	46.3	24
24	70.9	21	65.4	24	73.7	26	63.9	26	45.3	25
39	39.9	32	50.2	29	57.2	29	49.6	29	43.6	26
5	48.7	28	51.5	28	51.6	30	44.8	30	42.5	27
23	44.7	30	59.2	25	77.5	23	67.1	23	42.1	28
80	57.6	25	57.9	26	74.5	25	64.5	25	41	29
123	45.7	29	49.7	30	44.8	37	38.8	37	38.5	30
52	29.6	40	42.1	33	45.2	36	39.2	36	37.9	31
85	35.6	36	40.3	35	47.7	32	41.3	32	33	32
77	35.7	35	38.3	38	46.1	35	39.9	35	31.8	33
19	38	34	39	37	51.4	31	44.5	31	31	34
25	33.4	38	40.7	34	72.8	27	63.1	27	29.1	35
112	34.5	37	43.1	31	46.9	34	40.7	34	28.9	36
50	26	43	34.4	39	36.5	39	31.6	39	28.3	37
93	49.3	27	42.5	32	23.5	43	20.4	43	27.7	38
53	43.5	31	40	36	47.2	33	40.9	33	27.1	39
61	25.4	44	31.2	40	32.8	40	28.5	40	23.8	40
90	39.1	33	28.6	41	21	45	18.2	45	21.9	41
57	27.8	41	26.1	42	40.7	38	35.3	38	18.1	42
102	23.1	45	22.8	43	16.4	47	14.2	47	14.7	43
58	30.2	39	21.4	44	29.2	41	25.3	41	14.7	44
6	21.1	46	19.4	46	1.2	51	1	51	12.8	45

1 Table 1. (continue)

2											
	41	1.3	51	10.6	47	12.5	49	10.8	49	8.6	46
	82	5.4	48	8.9	49	24.5	42	21.3	42	8	47
	69	27	42	20.2	45	20.6	46	17.9	46	6.8	48
	42	5.6	47	6.1	50	16.2	48	14	48	6.8	49
	68	4.3	49	10.4	48	21.5	44	18.6	44	6	50
	104	-15.9	62	-7.7	54	-19.2	58	-16.6	58	1	51
	18	2.4	50	1.4	51	-1.8	52	-1.5	52	-2.4	52
	97	-3.6	54	-9.3	55	3.7	50	3.2	50	-2.5	53
	95	-8.5	56	-11.1	57	-22	61	-19.1	61	-6.6	54
	32	-10.3	57	-4.4	53	-12.6	53	-10.9	53	-7.3	55
	38	-0.9	52	-3.8	52	-14.9	55	-12.9	55	-8.2	56
	88	-10.8	58	-19.9	62	-31.8	64	-27.6	64	-8.5	57
	92	-13	59	-10.9	56	-18.1	57	-15.7	57	-9.2	58
	4	-15.1	61	-18.2	60	-16.5	56	-14.3	56	-12.2	59
	21	-37.6	73	-25.2	64	-19.7	59	-17.1	59	-15	60
	105	-14.4	60	-18.6	61	-13.2	54	-11.4	54	-15.7	61
	125	-26	64	-30.7	67	-30.7	63	-26.6	63	-16.7	62
	100	-33	69	-27.7	65	-21.5	60	-18.6	60	-18.4	63
	109	-2.5	53	-14	58	-54	75	-46.8	75	-18.6	64
	26	-27.9	65	-22.8	63	-33.7	65	-29.2	65	-19.4	65
	96	-7.5	55	-16.8	59	-30.4	62	-26.3	62	-19.4	66
	103	-28.5	66	-31.2	68	-34.7	66	-30.1	66	-21.4	67
	124	-25.8	63	-30.3	66	-50.5	73	-43.7	73	-25.6	68
	17	-33.8	70	-37.7	72	-61.4	78	-53.2	78	-27.1	69
	13	-50.7	81	-47.9	77	-47.1	71	-40.8	71	-28.4	70
	108	-29.6	67	-33.1	69	-36	67	-31.2	67	-29.5	71
	113	-50.2	79	-43.9	74	-43.9	70	-38	70	-30.2	72
	87	-31.7	68	-34.6	70	-40.3	69	-35	69	-30.4	73
	111	-36.7	72	-42.3	73	-54	74	-46.8	74	-32	74
	51	-39.7	74	-44.2	75	-70.7	87	-61.2	87	-32.5	75
	65	-42.2	75	-48.5	78	-54.8	76	-47.5	76	-34.1	76
	36	-52.6	82	-51.1	81	-61.7	79	-53.5	79	-37.2	77
	118	-45	77	-49.6	80	-66.8	83	-57.9	83	-38.1	78
	89	-34	71	-34.9	71	-37.9	68	-32.9	68	-39	79
	101	-42.2	76	-46	76	-62	80	-53.7	80	-40.9	80
	106	-48.1	78	-48.9	79	-64.7	81	-56	81	-41.3	81
	74	-60.4	86	-60.8	85	-70.2	86	-60.9	86	-41.4	82
	86	-62.4	88	-53.1	83	-66.2	82	-57.4	82	-43.2	83
	33	-65.1	89	-60.8	86	-70.1	85	-60.7	85	-44.3	84
	66	-59.1	84	-59.9	84	-67.2	84	-58.2	84	-45	85
	107	-50.2	80	-53	82	-48.6	72	-42.2	72	-45.4	86
	47	-60.2	85	-68.7	90	-101	94	-87.5	94	-45.5	87
	131	-66.2	91	-62.9	87	-78.2	90	-67.7	90	-46.6	88
	15	-59	83	-69.6	91	-75.5	89	-65.4	89	-50.1	89
	83	-78.4	93	-74.7	93	-80.1	91	-69.4	91	-52.4	90
	67	-65.7	90	-66.5	89	-75	88	-65	88	-53	91
	46	-61.1	87	-65.6	88	-57.1	77	-49.5	77	-54.1	92
	110	-77.5	92	-74.1	92	-95.1	92	-82.4	92	-57.5	93
	119	-87.6	94	-88.6	95	-104	95	-89.8	95	-64	94
	94	-90.1	95	-88.2	94	-108	98	-93.3	98	-65.6	95

1 **Table 1. (continue)**

2

14	-90.1	96	-96.4	97	-95.5	93	-82.7	93	-68.6	96
35	-94.5	98	-93.5	96	-113	100	-98.3	100	-68.7	97
91	-103	102	-100	101	-110	99	-95.1	99	-73.5	98
72	-99.9	101	-98.2	98	-126	103	-109	103	-74.1	99
64	-94.5	99	-99.9	100	-104	96	-90.5	96	-74.3	100
121	-91.4	97	-98.6	99	-107	97	-92.4	97	-75.4	101
116	-98.3	100	-103	102	-114	101	-98.8	101	-78.5	102
59	-127	104	-122	104	-130	104	-113	104	-81.9	103
114	-119	103	-115	103	-137	105	-119	105	-82.7	104
45	-135	106	-125	105	-124	102	-107	102	-90.6	105
115	-133	105	-140	106	-156	106	-135	106	-102	106
63	-172	107	-170	107	-191	107	-166	107	-120	107

3

4 ¹ *BV1, BV2 BV3, BV4 and BV5* are as defined in the text. Breeding values presented in the table

5 represent height gain in centimeter over the test population mean at plantation age of 20 years.

6

7